

Fixed-shift obstructions and optimized sieve constants for the Erdős–Straus conjecture

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Abstract

For a prime $p \equiv 1 \pmod{24}$ and a shift $c \equiv 3 \pmod{4}$, put $K_c = (p + c)/4$. The complete fixed-shift criterion expresses the existence of an Erdős–Straus decomposition whose first denominator is K_c in terms of two congruences between coprime divisors of K_c . We first prove that, for every fixed $c \equiv 3 \pmod{4}$, there are

$$\gg_c \frac{N}{(\log N)^{3/2}}$$

primes $p \leq N$, $p \equiv 1 \pmod{24}$, for which both congruences fail. The construction uses the principal form of discriminant $-c$ and the primitive shifted-prime theorem of Fuchs, Hsu, Rickards, Schindler and Stange. A signed divisor set, together with Kneser’s theorem, gives a general subgroup-concentration certificate for every failed shift.

We next evaluate exactly the two-parameter weight occurring in Dahan’s large-sieve argument:

$$\sum_{\substack{u, a \leq J \\ (u, a) = 1}} \frac{1}{\varphi(4ua)} = \frac{9}{2\pi^2} (\log J)^2 + \frac{9}{\pi^2} (\gamma + \Delta) \log J + O(1),$$

where $\Delta = \log 2/3 + \sum_{q > 2} \log q / ((q - 1)(q^2 + 1))$. Replacing the square window by the product window $ua \leq J^2$ doubles the leading weight at the same modulus budget. After a refined lower cutoff and the saturation argument, this gives

$$|E_H(N; J)| \ll N \exp \left\{ -(0.137873067 \dots + o(1)) (\log N)^{2/3} \right\}.$$

Finally, we prove that Dickson’s conjecture implies unbounded minimal successful shift, while one explicit instance of Bunyakovsky’s conjecture already gives infinitely many primes for which the shifts 3 and 7 both fail. These final implications are stated conditionally; none of the results proves the Erdős–Straus conjecture.

1 Introduction and statement of results

The Erdős–Straus conjecture asserts that for every integer $n \geq 2$ there are positive integers x, y, z such that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}. \tag{1}$$

Classical congruence reductions leave primes $p \equiv 1 \pmod{24}$ as the principal unresolved case; see Vaughan [7], Elsholtz–Tao [2], and Pomerance–Weingartner [5]. Dahan [1] introduced a complete divisor criterion at a fixed first denominator, proved quantitative exceptional-set estimates for a two-parameter Type II search, and constructed a two-sided obstruction at the shift $c = 7$. The present paper develops the fixed-shift obstruction for arbitrary c , determines the exact two-parameter weight, and optimizes the parameter region at fixed modulus budget.

Let $p \equiv 1 \pmod{24}$ be prime, let $c \equiv 3 \pmod{4}$ with $c < p$, and set

$$K_c = \frac{p+c}{4}. \quad (2)$$

We say that the shift c *fails at* p if neither branch of the complete fixed-shift criterion in Proposition 2.2 holds. Equivalently, no decomposition (1) has first denominator K_c . This is a statement about one prescribed first denominator, not a counterexample to (1). In the terminology of Dahan [1], such a prime is blind at the shift c .

Our first result is the following.

Theorem 1.1 (Every fixed shift has many failures). *Fix an integer $c \equiv 3 \pmod{4}$. The number of primes $p \leq N$ satisfying $p \equiv 1 \pmod{24}$ for which the shift c fails is*

$$\gg_c \frac{N}{(\log N)^{3/2}}.$$

More precisely, the lower bound is furnished by an explicit family of primes represented primitively by a shifted binary quadratic form.

The theorem extends the $c = 7$ construction of Dahan. It does not assert an asymptotic formula for the full failure set: higher-index subgroup certificates may produce additional failures.

The structural point is that the complete criterion can be written as a target-hitting problem in the unit group $(\mathbb{Z}/c\mathbb{Z})^\times$. If $K_c = \prod_j q_j^{e_j}$, define

$$\mathcal{B}_c(K_c) = \left\{ \prod_j q_j^{z_j} \pmod{c} : -e_j \leq z_j \leq e_j \right\}. \quad (3)$$

The shift succeeds precisely when this set contains -1 or $-p$. Kneser's theorem then shows that every failure forces almost all prime factors of K_c into one proper subgroup avoiding both targets. The precise result is Theorem 2.4 and Corollary 2.5. Quadratic-character obstructions, including those used in Theorem 1.1, are the index-two case of this statement.

The second main result evaluates the arithmetic weight left open after Dahan's Lemma 4.22.

Theorem 1.2 (Exact square-window weight). *As $J \rightarrow \infty$,*

$$W(J) := \sum_{\substack{u, a \leq J \\ (u, a) = 1}} \frac{1}{\varphi(4ua)} = C(\log J)^2 + 2C(\gamma + \Delta) \log J + O(1), \quad (4)$$

where

$$C = \frac{9}{2\pi^2}, \quad \Delta = \frac{\log 2}{3} + \sum_{q>2} \frac{\log q}{(q-1)(q^2+1)}. \quad (5)$$

The implied constant is absolute.

The factor at the prime 2 is essential: it equals $3/8$ after the two zeta factors are removed. The proof in Section 4 gives the full Euler product and the coefficient extraction through the linear term.

The square $u, a \leq J$ is not the maximal parameter set under the modulus condition $4ua \leq 4J^2$. The maximal set is the product window

$$\mathcal{P}_H(J) = \{(u, a) \in \mathbb{N}^2 : (u, a) = 1, ua \leq J^2\}. \quad (6)$$

Let $E_H(N; J)$ denote the primes $p \leq N$, $p \equiv 1 \pmod{24}$, for which Dahan's prime-witness form of the complete Type II criterion has no witness with $(u, a) \in \mathcal{P}_H(J)$.

Theorem 1.3 (Product-window exceptional set). *The product-window weight satisfies*

$$T(J) := \sum_{(u,a) \in \mathcal{P}_H(J)} \frac{1}{\varphi(4ua)} = \frac{9}{\pi^2} (\log J)^2 + \frac{18}{\pi^2} (\gamma + \Delta) \log J + O(1). \quad (7)$$

For fixed $\lambda \geq 1$ and $J = (\log N)^\lambda$,

$$|E_H(N; J)| \ll N \exp \left\{ - \left(\frac{9}{\pi^2} \lambda^2 + o(1) \right) (\log \log N)^3 \right\}. \quad (8)$$

There is also a choice

$$\log J \sim 0.549899512 (\log N)^{1/3}$$

for which

$$|E_H(N; J)| \ll N \exp \left\{ - (0.137873067 \dots + o(1)) (\log N)^{2/3} \right\}. \quad (9)$$

The last two estimates use the Bombieri–Vinogradov and large-sieve inputs of Dahan [1]; Section 5 verifies that the product window preserves their modulus and multiplicity hypotheses. The improvement is at fixed product, hence fixed modulus budget. An individual coordinate in (6) may be as large as J^2 .

We finally record two conditional consequences of the fixed-shift framework. For a prime $p \equiv 1 \pmod{24}$, let $c_{\min}(p)$ be the least $c \equiv 3 \pmod{4}$ for which the shift succeeds, and set $c_{\min}(p) = \infty$ if no such shift exists. Thus $c_{\min}$ is defined without assuming the Erdős–Straus conjecture.

Theorem 1.4 (Conditional on Dickson). *Assume Dickson’s conjecture for finite admissible systems of linear forms. For every $m \geq 1$ there are infinitely many primes $p \equiv 1 \pmod{24}$ for which all shifts $4a - 1$, $1 \leq a \leq m$, fail. Consequently*

$$\sup_{p \equiv 1 \pmod{24}} c_{\min}(p) = \infty$$

under Dickson’s conjecture.

Theorem 1.5 (Conditional on Bunyakovsky). *Assume Bunyakovsky’s conjecture for the polynomial*

$$P(t) = 576t^2 - 1536t + 1129.$$

Then infinitely many primes $p \equiv 1 \pmod{24}$ fail simultaneously at $c = 3$ and $c = 7$.

Theorem 1.4 is a general finite-shift construction. Theorem 1.5 shows that the first two shifts already reduce to one classical irreducible quadratic. Explicit affine lines for the first three and four shifts, and finite examples, are placed in Appendix A. All numerical checks are collected separately in Appendix B.

2 The fixed-shift criterion and subgroup concentration

We begin with the complete criterion in a form suited to additive combinatorics. The following elementary lemma is included to fix the coprimality conditions.

Lemma 2.1 (Binary unit fractions). *Let $A, M \in \mathbb{N}$ with $(A, M) = 1$. The equation*

$$\frac{A}{M} = \frac{1}{y} + \frac{1}{z}$$

has a solution in positive integers if and only if there are coprime divisors d_1, d_2 of M such that $d_1 + d_2 \equiv 0 \pmod{A}$.

Proof. Write $y = gY$, $z = gZ$, where $(Y, Z) = 1$. Then $(Y + Z)M = AgYZ$. Since $(Y + Z, YZ) = 1$, one has $YZ \mid M$; thus Y, Z are coprime divisors of M , and $A \mid Y + Z$ because $(A, M) = 1$. Conversely, given coprime $d_1, d_2 \mid M$ with $A \mid d_1 + d_2$, set

$$g = \frac{(d_1 + d_2)M}{Ad_1d_2}.$$

The coprimality of d_1, d_2 shows that $d_1d_2 \mid M$, so g is a positive integer and $1/(gd_1) + 1/(gd_2) = A/M$. $\square$

Applying the lemma to

$$\frac{4}{p} - \frac{1}{K_c} = \frac{c}{pK_c}$$

gives Dahan's complete fixed-shift criterion.

Proposition 2.2 (Complete fixed-shift criterion). *Let p be prime, $c \equiv 3 \pmod{4}$, $0 < c \leq 2p$, and $K_c = (p + c)/4 \in \mathbb{N}$. Then (1) has a decomposition with first denominator K_c if and only if there are coprime divisors d_1, d_2 of K_c satisfying at least one of*

$$d_1 + d_2 \equiv 0 \pmod{c}, \quad (10)$$

$$pd_1 + d_2 \equiv 0 \pmod{c}. \quad (11)$$

Proof. For $c \leq 2p$, both $(c, p) = 1$ and $(c, K_c) = 1$: a common divisor of c and K_c divides $4K_c - c = p$, and $c \neq p$ because their residue classes modulo 4 differ. Lemma 2.1 applied with $A = c$ and $M = pK_c$ gives coprime divisors u, v of pK_c with $u + v \equiv 0 \pmod{c}$. Since $p \nmid K_c$, each divisor of pK_c is d or pd with $d \mid K_c$. Coprimality excludes the case in which both u and v are divisible by p . The remaining cases are $(u, v) = (d_1, d_2)$ and $(u, v) = (pd_1, d_2)$, up to interchange. They give (10) and (11). Conversely, either congruence supplies the required pair u, v in Lemma 2.1. $\square$

Write $G_c = (\mathbb{Z}/c\mathbb{Z})^\times$. Since $(K_c, c) = 1$, every prime factor of K_c lies in G_c .

Proposition 2.3 (Signed divisor set). *If $K_c = \prod_{j=1}^r q_j^{e_j}$, then*

$$\mathcal{B}_c(K_c) = \{d_2d_1^{-1} \pmod{c} : d_1, d_2 \mid K_c, (d_1, d_2) = 1\}. \quad (12)$$

Consequently,

$$c \text{ succeeds at } p \iff \{-1, -p\} \cap \mathcal{B}_c(K_c) \neq \emptyset. \quad (13)$$

Proof. For coprime $d_1, d_2 \mid K_c$, each prime power $q_j^{e_j}$ is assigned to at most one of the two divisors. Hence $v_{q_j}(d_2) - v_{q_j}(d_1) \in [-e_j, e_j]$, which proves one containment in (12). Conversely, given exponents $-e_j \leq z_j \leq e_j$, assign $q_j^{z_j}$ to d_2 when $z_j > 0$, assign $q_j^{-z_j}$ to d_1 when $z_j < 0$, and assign no power when $z_j = 0$. The resulting divisors are coprime and realize the prescribed quotient. Finally, dividing (10) or (11) by d_1 gives the targets -1 and $-p$, respectively. $\square$

We now apply Kneser's theorem in the finite abelian group G_c ; see, for example, Tao–Vu [6, Chapter 5]. For a subgroup $H \leq G_c$ and $q \in G_c$, write

$$o_H(q) = \text{ord}_{G_c/H}(qH), \quad \nu_H = \#\{-H, -pH\} \in \{1, 2\}.$$

Theorem 2.4 (Target deficit). *If the shift c fails at p , there is a subgroup $H \leq G_c$ such that both target cosets are absent from $\mathcal{B}_c(K_c)/H$ and*

$$\sum_{j=1}^r (\min\{2e_j + 1, o_H(q_j)\} - 1) \leq [G_c : H] - \nu_H - 1. \quad (14)$$

Proof. For each j put

$$A_j = \{q_j^z : -e_j \leq z \leq e_j\} \subseteq G_c.$$

Then $\mathcal{B}_c(K_c) = A_1 \cdots A_r$. Let

$$H = \{h \in G_c : h\mathcal{B}_c(K_c) = \mathcal{B}_c(K_c)\}$$

be its stabilizer. The signed divisor set is a union of H -cosets. By failure and (13), the ν_H target cosets are absent, so

$$|\mathcal{B}_c(K_c)/H| \leq [G_c : H] - \nu_H. \quad (15)$$

Kneser's theorem, applied to $A_1, \dots, A_r$ and divided by $|H|$, gives

$$|\mathcal{B}_c(K_c)/H| \geq 1 + \sum_{j=1}^r (|A_j H/H| - 1). \quad (16)$$

In the cyclic subgroup generated by $q_j H$, the image of A_j is a consecutive symmetric interval of $2e_j + 1$ powers. It therefore has exactly

$$|A_j H/H| = \min\{2e_j + 1, o_H(q_j)\}$$

elements. Combining this identity with (15) and (16) proves (14). $\square$

Corollary 2.5 (Subgroup concentration). *If the shift c fails, there is a proper subgroup $H \leq G_c$ such that $-1, -p \notin H$ and*

$$\#\{q : q \mid K_c, q \bmod c \notin H\} \leq [G_c : H] - \nu_H - 1. \quad (17)$$

More strongly, the left side may be replaced by

$$\sum_{q^e \parallel K_c, q \notin H} (\min\{2e + 1, o_H(q)\} - 1).$$

Proof. Since $1 \in \mathcal{B}_c(K_c)$ and H stabilizes $\mathcal{B}_c(K_c)$, every $h \in H$ belongs to $\mathcal{B}_c(K_c)$. Thus failure implies $-1, -p \notin H$, and H is proper. If $q_j \notin H$, then $o_H(q_j) \geq 2$; because $e_j \geq 1$, its summand in (14) is at least one. Dropping the remaining nonnegative weight proves (17); retaining it gives the stronger assertion. $\square$

For $[G_c : H] = 2$, the right side of (17) is nonpositive. Thus every prime factor of K_c lies in H . This explains why an odd quadratic character supplies an extremal fixed-shift obstruction.

3 Failures at every fixed shift

Fix $c \equiv 3 \pmod{4}$ and put

$$r = \frac{c+1}{4}, \quad f_c(x, y) = x^2 + xy + ry^2. \quad (18)$$

This is a primitive positive-definite integral binary quadratic form with

$$\text{disc}(f_c) = 1 - 4r = -c. \quad (19)$$

Choose

$$B = B(c) = \begin{cases} 4, & c \equiv 3 \pmod{8}, c \not\equiv 2 \pmod{3}, \\ 16, & c \equiv 7 \pmod{8}, c \not\equiv 2 \pmod{3}, \\ 12, & c \equiv 3 \pmod{8}, c \equiv 2 \pmod{3}, \\ 48, & c \equiv 7 \pmod{8}, c \equiv 2 \pmod{3}. \end{cases} \quad (20)$$

We consider primes of the form

$$p = Bf_c(x, y) - c, \quad (x, y) = 1. \quad (21)$$

We use the following consequence of Fuchs–Hsu–Rickards–Schindler–Stange [3, Theorem 1.1]. If $f = ax^2 + bxy + dy^2$ is primitive, has nonsquare discriminant D , is positive definite for $D < 0$, and $(a, 2D) = 1$, while $A \neq 0$, $B_0 \in \mathbb{N}$, $(A, B_0) = 1$, and either $2 \mid AB_0$ or $D \not\equiv 5 \pmod{8}$, then the number of primes at most N represented primitively by $B_0f + A$ is $\gg N/(\log N)^{3/2}$. The same lower bound holds in a reduced class $\ell \pmod{m}$ provided

$$(m, 2DB_0) = 1, \quad (\ell - A, m) = 1. \quad (22)$$

Lemma 3.1. *The theorem just stated applies to f_c , $A = -c$, and $B_0 = B(c)$. If $c \equiv 1 \pmod{3}$, it also applies with the condition $p \equiv 1 \pmod{3}$. Consequently (21) produces*

$$\gg_c N(\log N)^{-3/2}$$

distinct primes $p \leq N$.

Proof. The form f_c is primitive and positive definite, its discriminant $-c$ is negative and hence nonsquare, and its leading coefficient is 1, so $(1, 2c) = 1$. The integer $B(c)$ is positive and even. Moreover $(c, B(c)) = 1$: the only possible odd prime factor of $B(c)$ is 3, and it occurs only when $c \equiv 2 \pmod{3}$. Thus $(A, B_0) = 1$ and $2 \mid AB_0$.

If $c \equiv 1 \pmod{3}$, take $m = 3$ and $\ell = 1$. In this case $3 \nmid cB$, so $(3, 2(-c)B) = 1$, while $(\ell - A, 3) = (1 + c, 3) = 1$. Hence (22) holds. In all other cases no additional progression is imposed. The cited theorem gives the claimed lower bound. $\square$

Lemma 3.2. *Every sufficiently large prime produced by Lemma 3.1 satisfies $p \equiv 1 \pmod{24}$.*

Proof. Suppose first that $c \equiv 3 \pmod{8}$. Then r is odd. A primitive pair (x, y) is nonzero modulo 2, and

$$f_c(x, y) \equiv x^2 + xy + y^2 \equiv 1 \pmod{2}.$$

Since $B \equiv 4 \pmod{8}$, equation (21) gives $p \equiv 4 - 3 \equiv 1 \pmod{8}$. If $c \equiv 7 \pmod{8}$, then $16 \mid B$, and $p \equiv -c \equiv 1 \pmod{8}$.

For the congruence modulo 3, the case $c \equiv 1 \pmod{3}$ was imposed in Lemma 3.1. If $c \equiv 2 \pmod{3}$, then $3 \mid B$ and $p \equiv -c \equiv 1 \pmod{3}$. Finally suppose $3 \mid c$. Since $4 \equiv 1 \pmod{3}$, one has $r \equiv 1 \pmod{3}$, and

$$f_c(x, y) \equiv x^2 + xy + y^2 \equiv (2x + y)^2 \pmod{3}.$$

Thus $f_c(x, y)$ is 0 or 1 modulo 3. Here $B \equiv 1 \pmod{3}$; a produced prime $p > 3$ cannot arise from the value 0, and hence $p \equiv 1 \pmod{3}$. Combining the congruences modulo 8 and 3 proves the lemma. $\square$

Let $\chi_c(n) = (n/c)$ be the Jacobi character modulo c . It is nontrivial because $c \equiv 3 \pmod{4}$ gives $\chi_c(-1) = -1$.

Lemma 3.3. *For every sufficiently large prime in (21), every prime divisor q of*

$$K_c = \frac{p+c}{4} = \frac{B}{4}f_c(x, y) \quad (23)$$

satisfies $\chi_c(q) = 1$.

Proof. The identity

$$4f_c(x, y) = (2x + y)^2 + cy^2 \quad (24)$$

controls prime factors of the form value. Let q be an odd prime with $q \mid f_c(x, y)$ and $q \nmid c$. If $q \mid y$, then (24) gives $q \mid 2x + y$, hence $q \mid x$, contrary to $(x, y) = 1$. Therefore y is invertible modulo q , and (24) gives

$$\left(\frac{-c}{q}\right) = 1.$$

Quadratic reciprocity for Jacobi symbols gives, because $(c-1)/2$ is odd,

$$\left(\frac{-c}{q}\right) = \left(\frac{-1}{q}\right) \left(\frac{c}{q}\right) = \left(\frac{-1}{q}\right) (-1)^{((c-1)/2)((q-1)/2)} \left(\frac{q}{c}\right) = \left(\frac{q}{c}\right) = \chi_c(q). \quad (25)$$

If $q \mid c$ and $q \mid f_c(x, y)$, then $q \mid p = Bf_c(x, y) - c$. For $p > c$, this is impossible. If $q = 2$, either $c \equiv 3 \pmod{8}$ and the form value is odd, or $c \equiv 7 \pmod{8}$ and $(2/c) = 1$.

It remains to treat the fixed factor $B/4 \in \{1, 4, 3, 12\}$. The prime 2 occurs only for $c \equiv 7 \pmod{8}$, when $(2/c) = 1$. The prime 3 occurs only for $c \equiv 2 \pmod{3}$. Together with $c \equiv 3 \pmod{4}$, this gives $c \equiv 11 \pmod{12}$, and the standard evaluation of the Jacobi symbol gives $(3/c) = 1$. Thus every prime factor in (23) lies in the kernel of χ_c . $\square$

Proof of Theorem 1.1. Let $H_c = \ker \chi_c$. Lemma 3.3 gives $\mathcal{B}_c(K_c) \subseteq H_c$. It also gives $\chi_c(K_c) = 1$. Since $p \equiv 4K_c \pmod{c}$ and 4 is a square,

$$\chi_c(p) = \chi_c(K_c) = 1.$$

On the other hand,

$$\chi_c(-1) = -1, \quad \chi_c(-p) = -1.$$

Neither target belongs to $\mathcal{B}_c(K_c)$, so the shift fails by Proposition 2.3. Lemmas 3.1 and 3.2 provide the stated number of primes in the required congruence class. $\square$

4 The exact two-parameter weight

We now prove Theorem 1.2. For $\Re s, \Re t > 0$, define

$$F(s, t) = \sum_{(u, a)=1} \frac{u^{-s} a^{-t}}{\varphi(4ua)}. \quad (26)$$

The coprimality condition means that an odd prime may occur in u or in a , but not both. Its local factor is

$$L_q(s, t) = 1 + \sum_{i \geq 1} \frac{q^{-is}}{\varphi(q^i)} + \sum_{j \geq 1} \frac{q^{-jt}}{\varphi(q^j)} \quad (q > 2). \quad (27)$$

At the prime 2, the factor 4 in the denominator gives

$$L_2(s, t) = \frac{1}{2} + \sum_{i \geq 1} 2^{-1-i-is} + \sum_{j \geq 1} 2^{-1-j-jt}. \quad (28)$$

Factor the two harmonic singularities as

$$F(s, t) = \zeta(1+s)\zeta(1+t)H(s, t). \quad (29)$$

Put $r_s = q^{-1-s}$ and $r_t = q^{-1-t}$. Direct summation of (27) gives, for $q > 2$,

$$H_q(s, t) = 1 + \frac{r_s + r_t}{q-1} - \frac{q+1}{q-1} r_s r_t. \quad (30)$$

Similarly, summing (28) gives

$$H_2(s, t) = \frac{1}{2}(1 - 2^{-1-s}2^{-1-t}). \quad (31)$$

For any fixed $\eta < 1/2$, the cancellation of the terms of order $1/q$ in (30) yields

$$H_q(s, t) = 1 + O(q^{-2+2\eta}) \quad (\Re s, \Re t \geq -\eta).$$

Hence the Euler product for H converges absolutely in a neighbourhood of $(0, 0)$.

At an odd prime,

$$H_q(0, 0) = \left(1 + \frac{2q}{(q-1)^2}\right) \left(1 - \frac{1}{q}\right)^2 = 1 + \frac{1}{q^2}. \quad (32)$$

At 2, equations (28) and (31) give

$$L_2(0, 0) = \frac{3}{2}, \quad H_2(0, 0) = \frac{3}{8}. \quad (33)$$

It follows that

$$\begin{aligned} H(0, 0) &= \frac{3}{8} \prod_{q>2} (1 + q^{-2}) \\ &= \frac{3}{8} \cdot \frac{4}{5} \frac{\zeta(2)}{\zeta(4)} = \frac{9}{2\pi^2} = C. \end{aligned} \quad (34)$$

To extract the summatory function, write

$$H(s, t) = \sum_{r, q \geq 1} h(r, q) r^{-s} q^{-t}.$$

Absolute convergence for $\Re s, \Re t \geq -\eta$ implies

$$\sum_{r, q} |h(r, q)| r^\eta q^\eta < \infty, \quad (35)$$

and in particular all fixed logarithmic moments converge absolutely. Comparing coefficients in (29), and denoting the harmonic sum by $H_X = \sum_{n \leq X} 1/n$, gives the exact identity

$$W(J) = \sum_{r \leq J} \sum_{q \leq J} h(r, q) H_{\lfloor J/r \rfloor} H_{\lfloor J/q \rfloor}. \quad (36)$$

Uniformly for $r \leq J$,

$$H_{\lfloor J/r \rfloor} = \log J - \log r + \gamma + O(r/J). \quad (37)$$

Because $r/J \leq (r/J)^\eta$ for $r \leq J$, equations (35)–(37) show that all terms containing a harmonic-sum error contribute $O(J^{-\eta} \log J)$. Extending the remaining r, q sums to infinity

costs $O(J^{-\eta}(\log J)^2)$. Thus (36) may be expanded coefficientwise through the constant term. The coefficient of $(\log J)^2$ is $H(0, 0)$. Since

$$H_s(0, 0) = - \sum_{r, q} h(r, q) \log r, \quad H_t(0, 0) = - \sum_{r, q} h(r, q) \log q,$$

the coefficient of $\log J$ is

$$2\gamma H(0, 0) + H_s(0, 0) + H_t(0, 0). \quad (38)$$

It remains to compute the two derivatives. Differentiating (30) at $(0, 0)$ gives

$$\frac{\partial}{\partial s} \log H_q(0, 0) = \frac{\log q}{(q-1)(q^2+1)} \quad (q > 2). \quad (39)$$

Equation (31) gives

$$\frac{\partial}{\partial s} \log H_2(0, 0) = \frac{\log 2}{3}. \quad (40)$$

The prime series is absolutely convergent. Therefore

$$\frac{H_s(0, 0)}{H(0, 0)} = \frac{H_t(0, 0)}{H(0, 0)} = \Delta.$$

Substitution into (38), together with (34), proves (4). Absolute second logarithmic moments in (35) bound the remaining terms by $O(1)$, completing the proof of Theorem 1.2.

The same calculation gives, uniformly as $U, A \rightarrow \infty$,

$$\sum_{\substack{u \leq U, a \leq A \\ (u, a) = 1}} \frac{1}{\varphi(4ua)} = C \log U \log A + C(\gamma + \Delta)(\log U + \log A) + O(1). \quad (41)$$

If $UA \leq J^2$, then $\log U \log A \leq (\log J)^2$, with equality in the leading term at $U = A = J$. Thus no rectangular deformation improves the square window at fixed maximum product.

In Dahan's Theorem 4.17, the weight enters linearly. Replacing the lower bound $3/(2\pi^2)$ used there by (34) gives

$$|E_2(N; (\log N)^\lambda)| \ll N \exp \left\{ - \left(\frac{9}{2\pi^2} \lambda^2 + o(1) \right) (\log \log N)^3 \right\}. \quad (42)$$

This corollary uses the unchanged remainder and large-sieve estimates in [1, Lemmas 4.19–4.28]; the arithmetic input replaced in that argument is exactly $W(J)$.

5 The product window and the optimized constant

Every integer n has exactly $2^{\omega(n)}$ ordered coprime factorizations $n = ua$: each prime power dividing n is assigned wholly to u or wholly to a . Therefore

$$T(J) = \sum_{n \leq J^2} \frac{2^{\omega(n)}}{\varphi(4n)}. \quad (43)$$

The corresponding Dirichlet series is the diagonal of (26):

$$\sum_{n \geq 1} \frac{2^{\omega(n)}}{\varphi(4n)} n^{-s} = F(s, s) = \zeta(1+s)^2 H(s, s). \quad (44)$$

The standard convolution identity

$$\sum_{m \leq X} \frac{\tau(m)}{m} = \frac{1}{2}(\log X)^2 + 2\gamma \log X + O(1)$$

and the same coefficient argument as in Section 4 yield

$$\sum_{n \leq X} \frac{2^{\omega(n)}}{\varphi(4n)} = \frac{C}{2}(\log X)^2 + 2C(\gamma + \Delta) \log X + O(1). \quad (45)$$

Taking $X = J^2$ proves (7); its leading constant is

$$\kappa_H = 2C = \frac{9}{\pi^2}. \quad (46)$$

We next verify the two changes required in Dahan's sieve. We use the standard Bombieri–Vinogradov and Brun–Titchmarsh estimates in the forms recorded, for example, in Iwaniec–Kowalski [4]. For an auxiliary prime ℓ , a pair (u, a) removes the residue class $-ua^{-1} \pmod{\ell}$ when $4ua \mid \ell + 1$.

Lemma 5.1 (Distinct residue classes). *If $\ell > J^4$ is prime, distinct pairs in $\mathcal{P}_H(J)$ remove distinct classes modulo ℓ .*

Proof. Suppose $(u, a), (v, b) \in \mathcal{P}_H(J)$ and $ua^{-1} \equiv vb^{-1} \pmod{\ell}$. Then $\ell \mid ub - va$. If $ub = va$, the two reduced fractions u/a and v/b are equal; since both pairs are coprime, they coincide. Otherwise

$$0 < |ub - va| < J^4,$$

because $u, a, v, b \leq J^2$. This contradicts $\ell > J^4$. $\square$

Lemma 5.2 (Uniform modulus range). *Fix $B_0 > 0$ and choose $K > 2B_0$. Put*

$$Y_0(J) = J^4(\log J)^K. \quad (47)$$

For all sufficiently large J and every $y \geq Y_0(J)$,

$$4J^2 \leq y^{1/2}(\log y)^{-B_0}. \quad (48)$$

Proof. The right side of (48) is eventually increasing, so it is enough to take $y = Y_0(J)$. There it equals

$$J^2(\log J)^{K/2}(4 \log J + K \log \log J)^{-B_0} \sim 4^{-B_0} J^2(\log J)^{K/2-B_0}.$$

Since $K/2 - B_0 > 0$, this exceeds $4J^2$ for large J . $\square$

For a prime ℓ , let

$$\rho_H(\ell) = \#\{(u, a) \in \mathcal{P}_H(J) : 4ua \mid \ell + 1\}$$

and define

$$S_H(J, z) = \sum_{Y_0(J) < \ell \leq z} \frac{\rho_H(\ell)}{\ell}. \quad (49)$$

Proposition 5.3 (Mean value). *In the range in which $J \rightarrow \infty$, $z > Y_0(J)$, and the Bombieri–Vinogradov estimate is uniform for moduli at most $4J^2$,*

$$S_H(J, z) \geq (\kappa_H + o(1))(\log J)^2 \log \frac{\log z}{\log Y_0(J)} - O(1/\log J). \quad (50)$$

Proof. Interchanging the pair and prime sums gives

$$S_H(J, z) = \sum_{(u,a) \in \mathcal{P}_H(J)} \sum_{\substack{Y_0(J) < \ell \leq z \\ \ell \equiv -1 \pmod{4ua}}} \frac{1}{\ell}.$$

Partial summation applied to

$$\pi(y; q, -1) = \frac{\text{Li}(y)}{\varphi(q)} + E(y; q, -1), \quad q = 4ua,$$

shows that the main term is

$$T(J)(\log \log z - \log \log Y_0(J)).$$

By (7), this is the main term in (50).

It remains to sum the errors. A modulus $4n$ occurs with multiplicity $2^{\omega(n)} \leq \tau(n)$. On a dyadic interval $[Y, 2Y]$, integration by parts bounds its total contribution by

$$\frac{1}{Y} \sum_{n \leq J^2} 2^{\omega(n)} \max_{w \leq 2Y} |E(w; 4n, -1)|.$$

By Cauchy–Schwarz, this is at most

$$\begin{aligned} & \frac{1}{Y} \left(\sum_{n \leq J^2} \max_{w \leq 2Y} |E(w; 4n, -1)| \right)^{1/2} \\ & \times \left(\sum_{n \leq J^2} 4^{\omega(n)} \max_{w \leq 2Y} |E(w; 4n, -1)| \right)^{1/2}. \end{aligned}$$

Lemma 5.2 permits Bombieri–Vinogradov in the first factor, giving $\ll_A (Y/(\log Y)^A)^{1/2}$. For the second, Brun–Titchmarsh and $4^{\omega(n)} \leq \tau(n)^2$ give

$$\sum_{n \leq J^2} \frac{\tau(n)^2}{\varphi(4n)} \ll (\log J)^{O(1)},$$

so the factor is $\ll Y^{1/2}(\log J)^{O(1)}$. Summing the dyadic intervals and taking A sufficiently large gives $O(1/\log J)$. This is the same divisor-weight estimate used in Dahan’s Lemma 4.23, and proves the proposition. $\square$

By Lemma 5.1, the sieve removes exactly $\rho_H(\ell)$ distinct classes once $\ell > Y_0(J)$. Moreover $\rho_H(\ell)$ is bounded above by the unrestricted divisor-pair count used in Dahan’s second-moment estimate. Thus the large-sieve argument [1, Theorem 4.28] applies with S_H in place of S : if

$$J \leq \exp((\log N)^{1/2}), \quad S_H(J, z) \log z \leq \frac{1}{2} \log N, \quad (51)$$

then

$$|E_H(N; J)| \ll N S_H(J, z) \exp(-S_H(J, z)). \quad (52)$$

For $J = (\log N)^\lambda$, Proposition 5.3 and the same saturation choice as in Dahan’s proof of Theorem 4.17 give

$$S_H(J, z) \geq (\kappa_H \lambda^2 + o(1))(\log \log N)^3.$$

Equation (8) follows from (52). As in Dahan's argument, primes $p \leq z$ below the sifting range contribute $O(z) = N^{o(1)}$ and are absorbed by the displayed bound.

We give the optimization leading to (9). Write $L = \log z$ and $\eta = \log J$. Since

$$\log Y_0(J) = 4\eta + K \log \eta = (4 + o(1))\eta,$$

Proposition 5.3 gives

$$S_H \geq (\kappa_H + o(1))\eta^2 \log \frac{L}{4\eta}. \quad (53)$$

For fixed L , the principal term on the right is maximal when

$$2 \log \frac{L}{4\eta} = 1, \quad \text{that is,} \quad \eta = \frac{L}{4\sqrt{e}}.$$

At this point

$$S_H \geq (\kappa_H + o(1)) \frac{L^2}{32e}. \quad (54)$$

For completeness, the saturation step remains valid while J varies. The entry of a prime at the upper endpoint changes S_H by $\rho_H(\ell)/\ell \ll_\varepsilon \ell^{-1+\varepsilon} = o(1)$. Moving the lower endpoint from $Y_0(J)$ to $Y_0(J+1)$ deletes at most

$$\ll_\varepsilon (Y_0(J+1) - Y_0(J))Y_0(J)^{-1+\varepsilon} \ll_\varepsilon J^{-1+4\varepsilon}(\log J)^{O(1)} = o(1)$$

for a fixed $\varepsilon < 1/4$; here the number of admissible divisor pairs of $\ell+1$ is $\ll_\varepsilon \ell^\varepsilon$. Finally, the weight of the pairs added when J increases is

$$\begin{aligned} T(J+1) - T(J) &= \sum_{J^2 < n \leq (J+1)^2} \frac{2^{\omega(n)}}{\varphi(4n)} \\ &\ll_\varepsilon J^{-1+2\varepsilon} \log \log J = o(1), \end{aligned} \quad (55)$$

using $2^{\omega(n)} \leq \tau(n) \ll_\varepsilon n^\varepsilon$ and $1/\varphi(n) \ll (\log \log n)/n$. Brun–Titchmarsh supplies at most an additional $O(\log \log z)$ factor, still $o((\log J)^2)$. Hence every jump of S_H is $o(S_H)$ in the optimized range, which is the hypothesis needed for Dahan's saturation argument.

Choose the largest L for which the second condition in (51) holds. The negligible-jump statement gives

$$S_H L = (1/2 + o(1)) \log N. \quad (56)$$

Combining (54) and (56) yields

$$L \leq (1 + o(1)) \left(\frac{16e}{\kappa_H} \log N \right)^{1/3}. \quad (57)$$

Therefore

$$S_H \geq \left\{ \frac{1}{2} \left(\frac{\kappa_H}{16e} \right)^{1/3} + o(1) \right\} (\log N)^{2/3}. \quad (58)$$

With $\kappa_H = 9/\pi^2$, the three constants are

$$\begin{aligned} \log z &\sim 3.626524091 (\log N)^{1/3}, \\ \log J &\sim 0.549899512 (\log N)^{1/3}, \\ \frac{1}{2} \left(\frac{9}{16e\pi^2} \right)^{1/3} &= 0.137873067 \dots \end{aligned}$$

Substitution in (52) proves (9) and completes the proof of Theorem 1.3.

6 Conditional consequences

6.1 Dickson's conjecture

For $a \geq 1$, put $c_a = 4a - 1$. Fix m and choose a positive integer b such that

$$6 \prod_{a=1}^m ac_a \mid b, \quad v_\ell(b) > \max_{1 \leq a \leq m} v_\ell(a) \quad (\ell \leq m+1 \text{ prime}). \quad (59)$$

Consider the $m+1$ linear forms

$$L_0(t) = 1 + 4bt, \quad L_a(t) = 1 + \frac{b}{a}t \quad (1 \leq a \leq m). \quad (60)$$

Lemma 6.1. *The forms in (60) are primitive, pairwise nonproportional, and admissible.*

Proof. Each form has constant coefficient 1, so it is primitive. Their positive linear coefficients are distinct, so they are pairwise nonproportional. Let ℓ be prime. If $\ell \leq m+1$, the strict valuation condition in (59) gives $\ell \mid b/a$ for every $a \leq m$; hence all forms are 1 modulo ℓ . If $\ell > m+1$, each form has at most one root modulo ℓ . The union of the root sets has size at most $m+1 < \ell$, so it does not cover all residue classes. This is admissibility. $\square$

Lemma 6.2. *If q is prime and $q \mid a$, then*

$$\left(\frac{q}{4a-1} \right) = 1.$$

Proof. For odd $q \mid a$, quadratic reciprocity and $4a-1 \equiv -1 \pmod{q}$ give

$$\left(\frac{q}{4a-1} \right) = (-1)^{((q-1)/2)((4a-2)/2)} \left(\frac{-1}{q} \right) = 1,$$

because $(4a-2)/2 = 2a-1$ is odd. If $q = 2$, then a is even and $4a-1 \equiv 7 \pmod{8}$, so $(2/(4a-1)) = 1$. $\square$

Proof of Theorem 1.4. By Dickson's conjecture and Lemma 6.1, infinitely many positive integers t make all the forms in (60) prime. Put $p = L_0(t)$. Since $6 \mid b$, one has $p \equiv 1 \pmod{24}$. For $c_a = 4a - 1$,

$$K_{c_a} = \frac{p + c_a}{4} = a + bt = aL_a(t). \quad (61)$$

Because $c_a \mid b$, one has $p \equiv 1 \pmod{c_a}$ and hence

$$K_{c_a} \equiv 4^{-1} \pmod{c_a}. \quad (62)$$

By Lemma 6.2, the Jacobi symbol of a modulo c_a is 1. Equations (61) and (62) therefore imply $(L_a(t)/c_a) = 1$. Since $L_a(t)$ is prime, every prime factor of K_{c_a} lies in the kernel of the Jacobi character modulo c_a .

The two targets coincide because $p \equiv 1 \pmod{c_a}$, and $(-1/c_a) = -1$ since $c_a \equiv 3 \pmod{4}$. Thus both targets lie outside the kernel, while the signed divisor set lies inside it. Every shift c_a , $1 \leq a \leq m$, fails. Since m is arbitrary, the asserted conditional unboundedness follows. $\square$

6.2 A Bunyakovsky reduction for two shifts

Put

$$f_1(x, y) = x^2 + xy + y^2, \quad f_2(u, v) = u^2 + uv + 2v^2.$$

The affine quadric

$$f_1(x, y) + 1 = 4f_2(u, v) \tag{63}$$

contains the integer line

$$(x, y, u, v) = (-19 + 12t, 6, -9 + 6t, 2). \tag{64}$$

Direct expansion gives

$$f_1(-19 + 12t, 6) = 144t^2 - 384t + 283, \tag{65}$$

$$f_2(-9 + 6t, 2) = 36t^2 - 96t + 71, \tag{66}$$

so (63) holds identically. Both pairs are primitive:

$$(-19 + 12t, 6) = 1, \quad (-9 + 6t, 2) = 1.$$

The associated prime polynomial is

$$P(t) = 4f_1(-19 + 12t, 6) - 3 = 576t^2 - 1536t + 1129 = 64(3t - 4)^2 + 105. \tag{67}$$

It has discriminant

$$(-1536)^2 - 4 \cdot 576 \cdot 1129 = -241920 < 0,$$

so it is irreducible over $\mathbb{Z}$ and positive. Its coefficients have gcd 1. Moreover

$$P(0) = 1129, \quad P(1) = 169, \quad (1129, 169) = 1,$$

so no prime divides every value of P . Finally $P(t) \equiv 1 \pmod{24}$.

Assume that $p = P(t)$ is prime. Equations (65) and (66) give

$$K_3 = f_1(-19 + 12t, 6), \quad K_7 = 4f_2(-9 + 6t, 2).$$

The primitive norm argument in Lemma 3.3 places every prime factor of K_3 in the quadratic kernel modulo 3 and every prime factor of K_7 in the quadratic kernel modulo 7. Also p has Jacobi symbol 1 modulo both 3 and 7, whereas -1 has symbol -1 . Proposition 2.3 shows that both shifts fail. Bunyakovsky's conjecture for the single polynomial (67) therefore proves Theorem 1.5.

7 Scope and remaining problem

Theorem 1.1 proves that no one prescribed shift can cover all hard primes. Theorem 1.3 gives a stronger upper bound for the exceptional set of the complete Type II search, but it does not show that this set is empty. The conditional results show that simultaneous failure of any fixed initial segment is compatible with standard prime-value conjectures; they do not supply an unconditional lower bound for even the intersection of the $c = 3$ and $c = 7$ failure sets.

In the language of Corollary 2.5, the missing unconditional input is a lower-bound theorem for a prime linear form while several parallel shifted values are constrained to lie, up to finitely many prime factors, in prescribed Frobenian multiplicative semigroups. For the first two shifts it is equivalently a prime-point problem on the affine quadric (63). Present one-variable correlation theorems and half-dimensional sieve estimates do not remove the relevant parity obstruction. This is the first analytic problem that must be solved to obtain unconditional progress toward unbounded minimal shift.

A Affine lines for three and four shifts

For $a \geq 1$, write

$$c_a = 4a - 1, \quad f_a(x, y) = x^2 + xy + ay^2.$$

Let $Z_a = (x_a, y_a)$ and $W_a = (u_a, v_a)$, and define the associated bilinear form

$$B_a((x, y), (u, v)) = 2xu + xv + yu + 2ayv.$$

If

$$f_a(W_a) = A, \quad B_a(Z_a, W_a) = L, \quad f_a(Z_a) = B_0 + a, \quad (68)$$

then

$$f_a(Z_a + sW_a) = As^2 + Ls + B_0 + a. \quad (69)$$

Writing $D_a = \det(Z_a, W_a)$, the binary Gram determinant is

$$4f_a(Z_a)f_a(W_a) - B_a(Z_a, W_a)^2 = (4a - 1)D_a^2. \quad (70)$$

Thus, with

$$C = 4AB_0 - L^2, \quad H = C + A,$$

equation (70) is equivalent to

$$D_a^2 = A + \frac{H}{4a - 1}. \quad (71)$$

The common prime candidate is

$$4(As^2 + Ls + B_0 + 1) - 3, \quad (72)$$

whose discriminant is $-16H$.

For the first three shifts, take

$$A = 163, \quad B_0 = 2, \quad L = 9, \quad H = 1386$$

and

a	Z_a	W_a	D_a
1	$(-2, 1)$	$(-3, -11)$	25
2	$(-2, 1)$	$(-7, -6)$	19
3	$(-2, 1)$	$(5, 6)$	-17

Substitution verifies (68) and gives

$$K_{4a-1}(s) = 163s^2 + 9s + a + 2, \quad P_3(s) = 652s^2 + 36s + 9. \quad (73)$$

The discriminant of P_3 is $-22176 = -16 \cdot 1386$. If a prime ℓ divides both coordinates of $Z_a + sW_a$, then $\ell \mid D_a$. For every prime divisor of $25 \cdot 19 \cdot 17$, the residue $s = 2$ avoids the unique bad class. Hence the restriction

$$s = 2 + Mt, \quad M = 3 \cdot 5 \cdot 17 \cdot 19,$$

makes all three pairs primitive and gives $P_3(s) \equiv 1 \pmod{24}$. The transformed quadratic is primitive, positive and irreducible; being always odd, it has no fixed prime divisor. More explicitly, $(P_3(0), P_3(1)) = (9, 697) = 1$, so P_3 has no fixed prime divisor. If $\ell \nmid M$, the map $t \mapsto 2 + Mt$ is a bijection modulo ℓ ; if $\ell \mid M$, then $P_3(2) = 2689 \not\equiv 0 \pmod{\ell}$. Hence

$P_3(2 + Mt)$ has no fixed prime divisor. Bunyakovsky's conjecture for this quadratic therefore implies infinitely many primes failing the first three shifts. The value $s = 2$ gives

$$P_3(2) = 2689, \quad c_{\min}(2689) = 15.$$

For the first four shifts, take

$$A = 2731, \quad B_0 = 56, \quad L = 405, \quad H = 450450$$

and

a	Z_a	W_a	D_a
1	$(-8, 1)$	$(-49, 55)$	-391
2	$(-8, 1)$	$(-19, -30)$	259
3	$(-8, 1)$	$(-31, 30)$	-209
4	$(-8, 1)$	$(-27, 26)$	-181

Again direct substitution gives

$$K_{4a-1}(s) = 2731s^2 + 405s + 56 + a, \quad P_4(s) = 10924s^2 + 1620s + 225. \quad (74)$$

Its discriminant is $-7207200 = -16 \cdot 450450$. The prime divisors of the four determinants are

$$7, 11, 17, 19, 23, 37, 181.$$

The class $s = 2$ avoids every bad primitivity class. Restricting

$$s = 2 + M_4t, \quad M_4 = 3 \cdot 7 \cdot 11 \cdot 17 \cdot 19 \cdot 23 \cdot 37 \cdot 181,$$

therefore produces a primitive, positive, irreducible quadratic with no fixed prime divisor and values 1 modulo 24. Indeed, $(P_4(0), P_4(1)) = (225, 12769) = 1$. For $\ell \nmid M_4$, the restricted parameter runs through every residue class modulo ℓ , while for $\ell \mid M_4$ one has $P_4(2) = 47161 \not\equiv 0 \pmod{\ell}$. Thus $P_4(2 + M_4t)$ has no fixed prime divisor. Bunyakovsky's conjecture for that polynomial implies infinitely many primes failing the first four shifts. At $s = 2$,

$$P_4(2) = 47161, \quad c_{\min}(47161) = 19.$$

For general m , this construction reduces the existence of a common affine line to the simultaneous-square system (71) for $1 \leq a \leq m$, with $H \neq 0$. We make no existence claim for arbitrary m .

B Numerical checks

The computations in this appendix are checks of identities and finite examples only; none is used in a proof above.

B.1 Arithmetic weights

Direct exact summation of $W(J)$ gives the following slowly convergent ratios.

J	$W(J)/(\log J)^2$
100	0.6279474967
300	0.5950299810
1000	0.5711416342
3000	0.5555895264

They approach the proved constant $9/(2\pi^2) = 0.4559453264\dots$. For the product window, subtracting the two terms in (7) gives residuals

J	$T(J)/(\log J)^2$	two-term residual
100	1.2651142845	0.01295727
300	1.1969484147	0.01178495
1000	1.1472139536	0.01184315

The limiting leading constant is $9/\pi^2 = 0.9118906528\dots$

B.2 Fixed-shift failure counts

The complete signed-divisor test was applied to the 82887 primes $p \leq 10^7$, $p \equiv 1 \pmod{24}$. The number for which both branches fail is:

c	count	c	count
3	35750	19	46440
7	18732	23	17255
11	25242	27	49582
15	20414	31	34907

These are counts for the full failure sets, not only the quadratic-form subfamilies constructed in Section 3.

A separate complete two-branch computation tested all 719781 primes $p < 10^8$, $p \equiv 1 \pmod{24}$, through shifts $c \leq 503$. The largest observed minimal shift was

$$c_{\min}(8803369) = 107,$$

and every tested prime succeeded by $c = 107$. This finite statement does not imply a uniform bound.

B.3 Exact conditional-family examples

For the Dickson construction with $m = 4$, the choice $b = 27720$, $t = 111$ gives

$$p = 12307681$$

and

$$K_3 = 3076921, \quad K_7 = 2 \cdot 1538461, \quad K_{11} = 3 \cdot 1025641, \quad K_{15} = 4 \cdot 769231.$$

The displayed cofactors are prime; the complete criterion fails at 3, 7, 11, 15 and first succeeds at 19. The examples $P_3(2) = 2689$ and $P_4(2) = 47161$ were recorded in Appendix A.

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